



8587 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Thushara Pillai (PI)	Massachusetts Institute of Technology
Dr. Jens Kauffmann (CoI) (CoPI)	Massachusetts Institute of Technology
Shuo Kong (CoI) (CoPI)	University of Arizona
Dr. Robert Gutermuth (CoI) (CoPI)	University of Massachusetts - Amherst
Dr. Tracy L. Huard (CoI) (CoPI)	University of Maryland
Dr. Neal J. Evans II (CoI)	University of Texas at Austin
Prof. Steven Longmore (CoI) (ESA Member)	Liverpool John Moores University
Dr. Hector G. Arce (CoI)	Yale University
Dr. Klaus M. Pontoppidan (CoI)	Jet Propulsion Laboratory
Dr. James Urquhart (CoI) (ESA Member)	University of Kent
Ms. Sudeshna Patra (CoI)	Indian Institute of Science Education and Research Tirupati
Dr. Jessy Jose (CoI)	Indian Institute of Science Education and Research Tirupati
Dr. Christoph Federrath (CoI)	Australian National University
Mr. Pacome Esteve (CoI) (ESA Member)	Institut d'Astrophysique Spatiale
Dr. Stella Offner (CoI) (CoPI)	University of Texas at Austin
Dr. Henrik Beuther (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Rebecca Houghton (CoI) (ESA Member)	Liverpool John Moores University

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Obs. G28-Center				
	1	NIRCam G28	NIRCam Imaging	(1) G28.333+0.063-CENTER
	2	MIRI G28	MIRI Imaging	(1) G28.333+0.063-CENTER

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Obs. G28-Reference				
	3	NIRCam G28 - Reference	NIRCam Imaging	(2) G28.333+0.063-REFERENCE
	4	MIRI G28 - Reference	MIRI Imaging	(2) G28.333+0.063-REFERENCE

ABSTRACT

Most star formation in the universe takes place in high-mass clusters. Their formation and evolution are thus fundamental to understanding how galaxies evolve. High-mass infrared dark clouds (IRDCs), extremely massive and dense filaments at the heart of some giant molecular clouds, are progenitors of high-mass clusters. Large distances and enormous extinction in the infrared have hindered any meaningful census of young stellar objects (YSOs) in IRDCs using infrared techniques, which provide the only means to directly sense the young stars. Here, we propose to use JWST to provide a complete characterization of YSOs down to 0.1 Lsun in one of the most well-studied IRDCs: G28.34+0.06, the “Dragon”. This JWST program enables several experiments crucial for our understanding of high-mass cluster formation (HMCF) and extragalactic star formation relations. It will provide the first direct census of YSOs and star formation rate in a high-mass IRDC. We can study the emergence and evolution of stellar populations in a cloud that has not yet been substantially perturbed by star formation, and is thus representative of the initial conditions. Since all known evolutionary phases of HMCF are embedded in the Dragon, we can place stringent constraints on the main models of HMCF. Further, star formation (SF) rate predictions from different model SF relations differ by an order of magnitude. This permits using the Dragon to decisively differentiate between different star formation relations, thus critically advancing our understanding of the processes governing galactic and extragalactic star formation.

OBSERVING DESCRIPTION

Relatively straightforward mosaicing observations with NIRCam and MIRI, with dithers and other parameters optimized to deliver a relatively homogeneous coverage with limited overheads. The observations also target a separate "control field" that is used to characterize the populations of foreground and background stars onto which our targets are imposed.

Proposal 8587 - Targets - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	G28.333+0.063-CENTER	RA: 18 42 52.7625 (280.7198437d) Dec: -04 02 5.77 (-4.03494d) Equinox: J2000		
	Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]				
Fixed Targets	(2)	G28.333+0.063-REFERENCE	RA: 18 41 54.1485 (280.4756187d) Dec: -03 59 11.12 (-3.98642d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 8587 - Observation 1 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Observation	Proposal 8587, Observation 1: NIRCam G28										Tue Nov 25 20:00:12 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(NIRCam G28 (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	G28.333+0.063-CENTER	RA: 18 42 52.7625 (280.7198437d) Dec: -04 02 5.77 (-4.03494d) Equinox: J2000								
	Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module gap (large extended source)				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	2	1	10.0		10.0		0.0		0.0		DEFAULT
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		4TIGHT		SMALL-GRID-DITHER				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F162M+F150W2	F360M	BRIGHT1	5	1	8	8	773.047		
	2	F210M	F480M	BRIGHT1	5	1	8	8	773.047		
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA										

Proposal 8587 - Observation 2 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Observation	Proposal 8587, Observation 2: MIRI G28										Tue Nov 25 20:00:12 GMT 2025
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	G28.333+0.063-CENTER	RA: 18 42 52.7625 (280.7198437d) Dec: -04 02 5.77 (-4.03494d) Equinox: J2000								
	Comments: Category=Stellar Cluster Description=[Protoclusters, Young star clusters]										
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	3	6	10.0		17.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	4		5	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	15	5	1	Dither 1	4	20	876.913	
	2	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	

Proposal 8587 - Observation 2 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Special Requirements	Group Visits within 53.0 Days Visits Same PA
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Proposal 8587 - Observation 3 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Observation	Proposal 8587, Observation 3: NIRCam G28 - Reference									Tue Nov 25 20:00:12 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRCam Imaging									
Diagnostics	(NIRCam G28 - Reference (Obs 3)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous	
	(2)	G28.333+0.063-REFERENCE	RA: 18 41 54.1485 (280.4756187d) Dec: -03 59 11.12 (-3.98642d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i>							
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		4TIGHT		SMALL-GRID-DITHER				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F162M+F150W2	F360M	BRIGHT1	5	1	8	8	773.047	
	2	F210M	F480M	BRIGHT1	5	1	8	8	773.047	

Proposal 8587 - Observation 4 - Star Formation in a High-Mass IRDC: Suppressed or Veiled?

Observation	Proposal 8587, Observation 4: MIRI G28 - Reference										Tue Nov 25 20:00:12 GMT 2025
	Diagnostic Status: Warning										
Observing Template: MIRI Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 4:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	G28.333+0.063-REFERENCE	RA: 18 41 54.1485 (280.4756187d) Dec: -03 59 11.12 (-3.98642d) Equinox: J2000								
Comments: Category=Calibration Description=[Telescope/sky background]											
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	2	2	10.0		17.0		0.0		0.0		DEFAULT
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	4		5	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	15	5	1	Dither 1	4	20	876.913	
	2	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	
Special Requirements	Group Visits within 53.0 Days Visits Same PA										